Dr. Yonatan Ashenafi

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Education

Rensselaer Polytechnic Institute (RPI)- Troy, NY

- Doctoral degree in mathematics: May-2021

Thesis: Stochastic Hydrodynamics of Colonial Microswimmers

Rensselaer Polytechnic Institute (RPI)-- Troy, NY

- Master's degree in applied mathematics: August 2016-December 2018

Dordt University- Sioux Center, IA

- Bachelor of Arts in mathematics: August 2012-May 2016

Experience

Technical Fellow Intern, *General Electric (GE) Research* *October – December 2020*

GE Global Research Center, Niskayuna, New York

- Collaborating with the probabilistic design team on multiple projects on experimental design and additive manufacturing.
- Using Python for programming for Reinforcement Learning tasks.

Postdoctoral Fellow, *Worcester Polytechnic Institute (WPI)* *August 2023 –Present*

Worcester Polytechnic Institute, Worcester, Massachusetts

- Conducting research on intracellular transport processes
- We are using data from our experimentalist collaborators at two other universities.
- Teaching Multivariable Calculus, Calculus 1, Calculus 2, and Ordinary Differential Equations.

Postdoctoral Fellow, *University of Alberta* *September 2021 – May 2023*

University of Alberta, Edmonton, Alberta

- Conducting research on the mechanisms of motion of important algae called diatoms.
- We used video data from experiments to construct and analyze our model
- Teaching Calculus 1 for life sciences and Calculus 2 for physical sciences.

Research Assistant, Dordt College *June-August 2014*

Undergraduate Research, Dordt College-Sioux Center, Iowa

- We utilized tabulated data about gene expression levels to construct a Gaussian mixture model.
- Trained in the statistical software, R, and utilized it to analyze data on the genetic expression of bacteria in various mediums.

Tutor, *Dordt College Library* *September 2013– May 2016*

Academic Skills Center, Dordt College-Sioux Center, Iowa

- Tutored students who requested assistance in the following classes: College Algebra, Calculus 1, Calculus 2, Multivariable Calculus, Linear Algebra, and Discrete Structures.
- Cooperated and communicated effectively with tutees, faculty, and staff.

Services and Activities

- Proposal Reviewer, National Science Foundation (NSF), 2025–present.
- Volunteer grader for the 2024 annual math meet at WPI.
- Volunteer judge for the 2022 annual undergraduate research symposium at the University of Alberta.
- Treasurerof Black Graduate Students Association at RPI for 2020. I managed the financing of our events during a challenging pandemic period.
- Presented at Ethiopian Mathematics Professional Association monthly seminar in July 2025.
- Presented at Biophysical Society annual meeting 2025, Biology and Medicine Through Mathematics (BAMM) 2024, Biological Fluid Mechanics mini symposium of SIAM Life Science 2021, University of Alberta's Mathematical Biology Seminar Series Fall 2021, Mathematical Biosciences Institute (MBI) Mathematical and Computational Methods in Biology Workshop 2020, and Conference on Multiscale modeling in Biology at University of Minnesota Summer 2019.

- Organized a mini symposium on the role of noise and asymmetry in microscopic life at the Canadian Applied and Industrial Mathematical Society (CAIMS) 2022
- Mathematical modeler for the Mathematical Problems in Industry (MPI) workshop at University of Vermont in 2024 and at RPI in 2017. Some of the work is publicly available.
- Participant at Systemic Initiative for Modeling Investigations & Opportunities with Differential Equations (SIMIODE) Spring 2024 Webinars

Awards

- Di Prima summer research fellowship for graduate research in the summer of 2018.
- Bill and Nancy Siegmann Applied Mathematical Modeling Prize for my thesis in May 2021.
- 2022 ASME CIE Advanced Modeling and Simulation Best Paper Award.

Professional Membership

- Society of Industrial and Applied Mathematics (SIAM).

Publications

- Yonatan Ashenafi and Peter R. Kramer. “Statistical Mobility of Multicellular Colonies of Flagellated Swimming Cells”. Bulletin of Mathematical Biology (2024).
- Yonatan Ashenafi. “Stability and Spatial Autocorrelations of Suspensions of Microswimmers with Heterogeneous Spin”. Physics of Fluids (2022).
- Yonatan Ashenafi, Piyush Pandita, and Sayan Ghosh. “Reinforcement learning based sequential batch-sampling for Bayesian optimal experimental design” Journal of Mechanical Design (2021).
- Zheng, Peng et al. “Cooperative motility, force generation and mechanosensing in a foraging non-photosynthetic diatom.” Open biology vol. 13,10 (2023): 230148. doi:10.1098/rsob.230148
- Disselkoen, Craig, et al. "A Bayesian framework for the classification of microbial gene activity states." *Frontiers in microbiology* 7 (2016): 1191.
- Yonatan Ashenafi and Peter R. Kramer. “Asymptotic Analysis of Kinesis and Taxis for Colonial Protozoa”. Under review.
- Yonatan Ashenafi and Jay Newby. “Adhesion and Geometry Govern Mobility in Raphid Diatoms”. In preparation.
- Yonatan Ashenafi and Sam Walcott. “Simulation-Based Inference of Hidden Motor States in Intracellular Cargo Transport”. In preparation.

Programming Skills Python (Pandas, NumPy, SciPy, TensorFlow, scikit-learn), MATLAB, C++, R